CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 6: Sets

Topic-wise short notes with solved examples

Sets and their notation, types of sets, subsets, the operations of union, intersection, difference and complement, Venn diagrams and the counting formula for two sets.

1 Sets and elements

- A **set** is a **well-defined** collection of distinct objects. "Well-defined" means it is clear whether any given object belongs to it or not.
- The objects are the **elements** (members). $a \in A$ means "a belongs to A"; $a \notin A$ means it does not.
- "The vowels of the English alphabet" is a set. "Good cricketers" is not, since "good" is not well-defined.
- Sets are named by capital letters. Elements are not repeated, and their order does not matter.
- **Roster (tabular) form**: list the elements in braces. $A = \{2, 4, 6, 8\}$.
- **Set-builder (rule) form**: state the property. $A = \{x : x \text{ is an even natural number less than } 10\}$.
- **Cardinal number** $n(A)$: the number of elements in A.

Example 1.1 Write $B = \{x : x \in N, 3 \leq x < 8\}$ in roster form and find $n(B)$.

SOLUTION

$B = \{3, 4, 5, 6, 7\}$; $n(B) = 5$

Example 1.2 Write $\{1, 4, 9, 16, 25\}$ in set-builder form.

SOLUTION

$\{x : x = n^2, n \in N, n \leq 5\}$

Example 1.3 Write the set of letters of the word MATHEMATICS.

SOLUTION

$\{M, A, T, H, E, I, C, S\}$, **with 8 elements**

2 Types of sets

Type	Meaning	Example
Empty (null) set, ϕ or $\{\}$	No elements	Even prime numbers greater than 2
Singleton set	Exactly one element	$\{0\}$
Finite set	A countable number of elements	Days of the week
Infinite set	Elements cannot be counted to an end	$N = \{1, 2, 3, \ldots\}$
Equal sets	Exactly the same elements	$\{1, 2, 3\}$ and $\{3, 1, 2\}$
Equivalent sets	The same number of elements	$\{a, b, c\}$ and $\{1, 2, 3\}$
Disjoint sets	No common element	$\{1, 3\}$ and $\{2, 4\}$
Overlapping sets	At least one common element	$\{1, 2, 3\}$ and $\{3, 4\}$

- $\{0\}$ and $\{\phi\}$ are **not** empty sets; each has one element.
- Equal sets are always equivalent. Equivalent sets need not be equal.
- The **universal set** ξ (or U) contains all the elements under discussion.

3 Subsets

- A is a **subset** of B, written $A \subseteq B$, if every element of A is in B.
- A is a **proper subset** of B, written $A \subset B$, if $A \subseteq B$ and $A \neq B$.
- Every set is a subset of itself. The empty set is a subset of every set.
- A set with n elements has 2^n subsets and $2^n - 1$ proper subsets.

Example 3.1 Write all the subsets of $\{a, b, c\}$.

SOLUTION

ϕ, $\{a\}$, $\{b\}$, $\{c\}$, $\{a, b\}$, $\{a, c\}$, $\{b, c\}$, $\{a, b, c\}$: **8 subsets**

Example 3.2 How many subsets and proper subsets does a set with 5 elements have?

SOLUTION

$2^5 = 32$ **subsets and 31 proper subsets**

4 Operations on sets

Operation	Symbol	Meaning
Union	$A \cup B$	Elements in A or in B or in both
Intersection	$A \cap B$	Elements in both A and B
Difference	$A - B$	Elements in A but not in B
Complement	A'	Elements of ξ not in A

- $A \cup B = B \cup A$ and $A \cap B = B \cap A$, but $A - B$ is usually different from $B - A$.
- $A \cup A' = \xi$, $A \cap A' = \phi$ and $(A')' = A$.
- For disjoint sets, $A \cap B = \phi$.
- $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$.

Example 4.1 Let $\xi = \{1, 2, \ldots, 10\}$, $A = \{1, 2, 3, 4, 5\}$ and $B = \{4, 5, 6, 7\}$. Find $A \cup B$, $A \cap B$, $A - B$, $B - A$ and A'.

SOLUTION

$A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$; $A \cap B = \{4, 5\}$; $A - B = \{1, 2, 3\}$; $B - A = \{6, 7\}$; $A' = \{6, 7, 8, 9, 10\}$

Example 4.2 For the same sets, verify that $(A \cup B)' = A' \cap B'$.

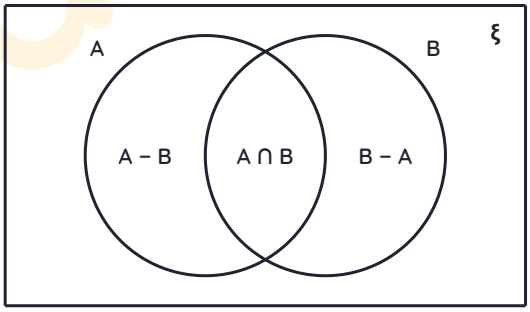
SOLUTION

$(A \cup B)' = \{8, 9, 10\}$. $A' = \{6, 7, 8, 9, 10\}$ and $B' = \{1, 2, 3, 8, 9, 10\}$.

$A' \cap B' = \{8, 9, 10\}$, **the same set**

5 Venn diagrams

- A **Venn diagram** shows the universal set as a rectangle and other sets as closed curves (usually circles) inside it.
- Overlapping circles show common elements. Separate circles show disjoint sets. A circle inside another shows a subset.



The three regions of two overlapping sets

6 Counting elements

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

- For disjoint sets, $n(A \cup B) = n(A) + n(B)$.
- $n(A - B) = n(A) - n(A \cap B)$ and $n(A') = n(\xi) - n(A)$.
- "Only A" means $A - B$. "Neither" means $(A \cup B)'$.

Example 6.1 $n(A) = 20$, $n(B) = 16$ and $n(A \cap B) = 7$. Find $n(A \cup B)$ and $n(A - B)$.

SOLUTION

$20 + 16 - 7$ and $20 - 7$

29 and 13

Example 6.2 In a class of 40 students, 25 play cricket, 18 play football and 8 play both. How many play neither?

SOLUTION

$n(C \cup F) = 25 + 18 - 8 = 35$

$40 - 35 = 5$

Example 6.3 In a group of 60 people, each speaks Hindi or English. 45 speak Hindi and 30 speak English. How many speak both? How many speak only Hindi?

SOLUTION

$60 = 45 + 30 - n(H \cap E)$, so $n(H \cap E) = 15$.

15 speak both; 30 speak only Hindi

7 Common mistakes

- Calling $\{0\}$ or $\{\phi\}$ an empty set.
- Repeating elements, as in writing the letters of a word.
- Confusing $\in$ (element of) with $\subseteq$ (subset of): $2 \in \{1, 2\}$ but $\{2\} \subseteq \{1, 2\}$.
- Writing $A - B = B - A$.
- Adding $n(A)$ and $n(B)$ without subtracting the common elements.
- Forgetting the empty set and the set itself when listing subsets.